Business Problem Overview and Solution Approach

Overview:

- Refurbished cell phone market has grown considerably over the past decade. Refurbished or used phones are cost effective, environmental impact is less and they also come with good insurance or warranties.
- ReCell is a startup company in the refurbished market. Pricing the cell phones in this market is challenging, as it is a balance of winning customers Vs having a good profit margin.
- ReCell wants to correctly price the refurbished cell phones using a data driven solution and identify the factors that influence the decisions.

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ReCell project

Supervised Learning Foundations

07/15/2025

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- Data Preprocessing
- Model Performance Summary
- Appendix

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Executive Summary

Actionable insights & recommendations

- The linear regression model is able to explain approximately 84% of the variation in the data. This indicates that the model is good for prediction as well as inference.
- Originally expensive devices have a good refurbished price too.
- There are many predictors of price of refurbished devices:
    - If the RAM increases by one unit, price increases by 0.0217 units..
    - If the resolution for main camera increases by one unit, then its price increases by 0.0212 units.
    - If the resolution for selfie camera increases by one unit, then its price increases by 0.0140 units, all other variables held constant.
    - If the weight of the device increases by one unit, then its price increases by 0.0017 units, all other variables held constant. Weight and screen size may be correlated.
    - Years since release has a slight negative correlation with the price.
- ReCell may advertise main camera and selfie camera features which can help with more resale of the devices.
- ReCell may focus on selling devices with more memory, good cameras and bigger screen sizes. These seem to be important features of the refurbished devices.

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Solution approach

- The solution involves analyzing the many characteristics of used cell phones, such as brand, memory, screen size etc to understand how they influence pricing.
- Univariate, bivariate analysis and building a linear regression model to predict the correct price can provide insights into which factors out of those are important.
- The model can suggest changes in the business strategy and can also provide a correct price for the refurbished cell phones.

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EDA Results

The data contains the different attributes of used/refurbished phones and tablets.

Data Dictionary

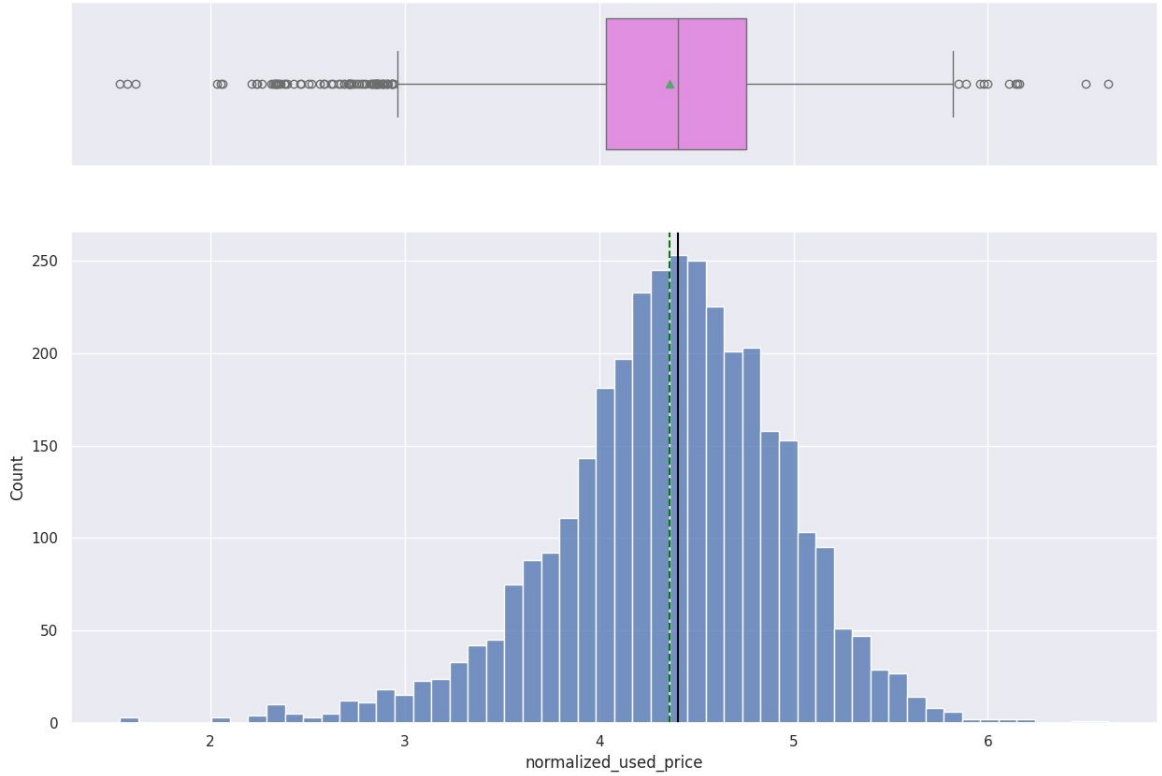
- brand_name: Name of manufacturing brand
- os: OS on which the device runs
- screen_size: Size of the screen in cm
- 4g: Whether 4G is available or not
- 5g: Whether 5G is available or not
- main_camera_mp: Resolution of the rear camera in megapixels
- selfie_camera_mp: Resolution of the front camera in megapixels
- int_memory: Amount of internal memory (ROM) in GB
- ram: Amount of RAM in GB
- battery: Energy capacity of the device battery in mAh
- weight: Weight of the device in grams
- release_year: Year when the device model was released
- days_used: Number of days the used/refurbished device has been used
- normalized_new_price: Normalized price of a new device of the same model in euros
- normalized_used_price: Normalized price of the used/refurbished device in euros

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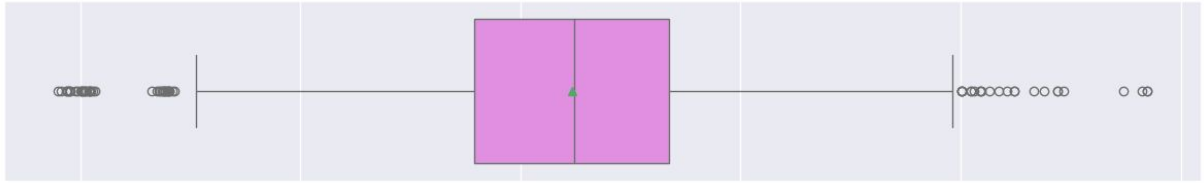
EDA Results - Univariate

Data for normalized used price is n
distributed.

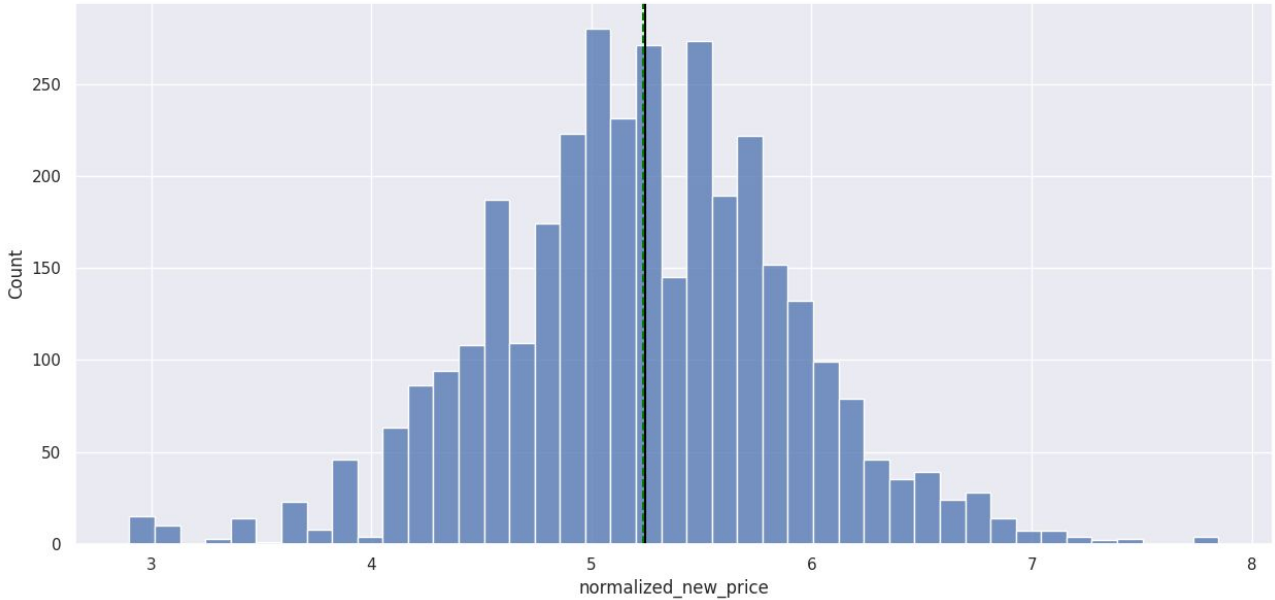


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EDA Results



Normalized new price data is normally distributed.



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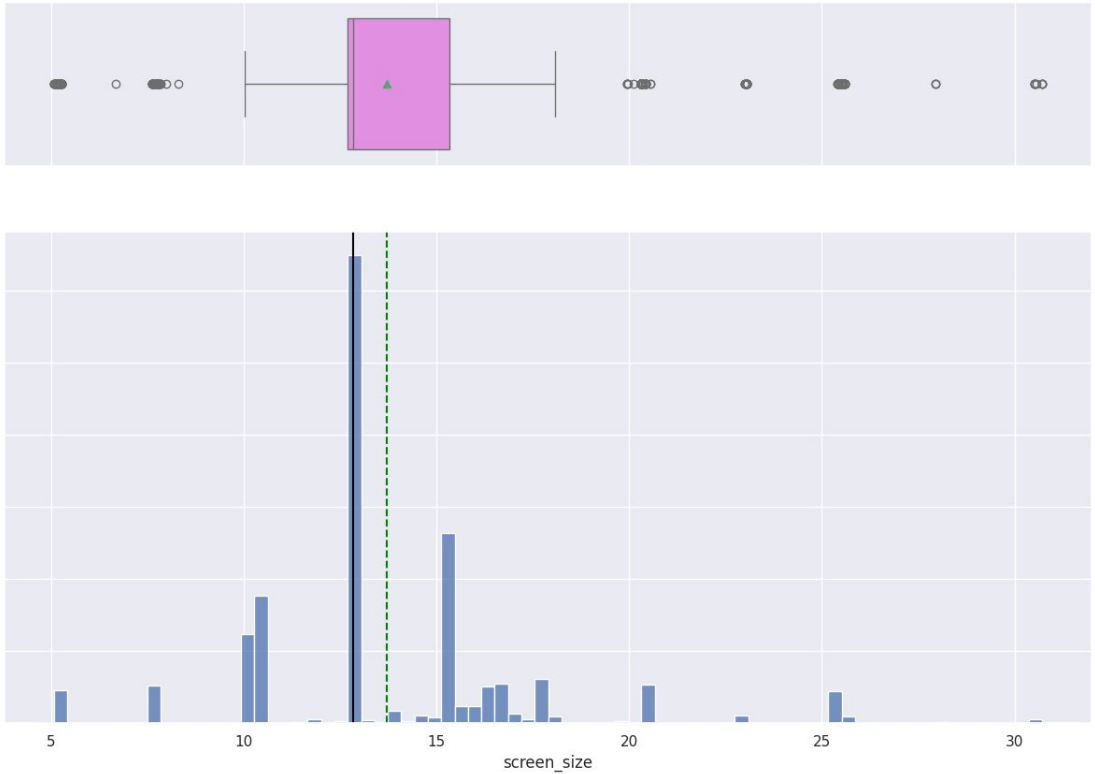
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EDA Results - Univariate- Screen size

Most screen sizes are concentrated into a few data points-

12, 15, 10

Data looks almost normalised.

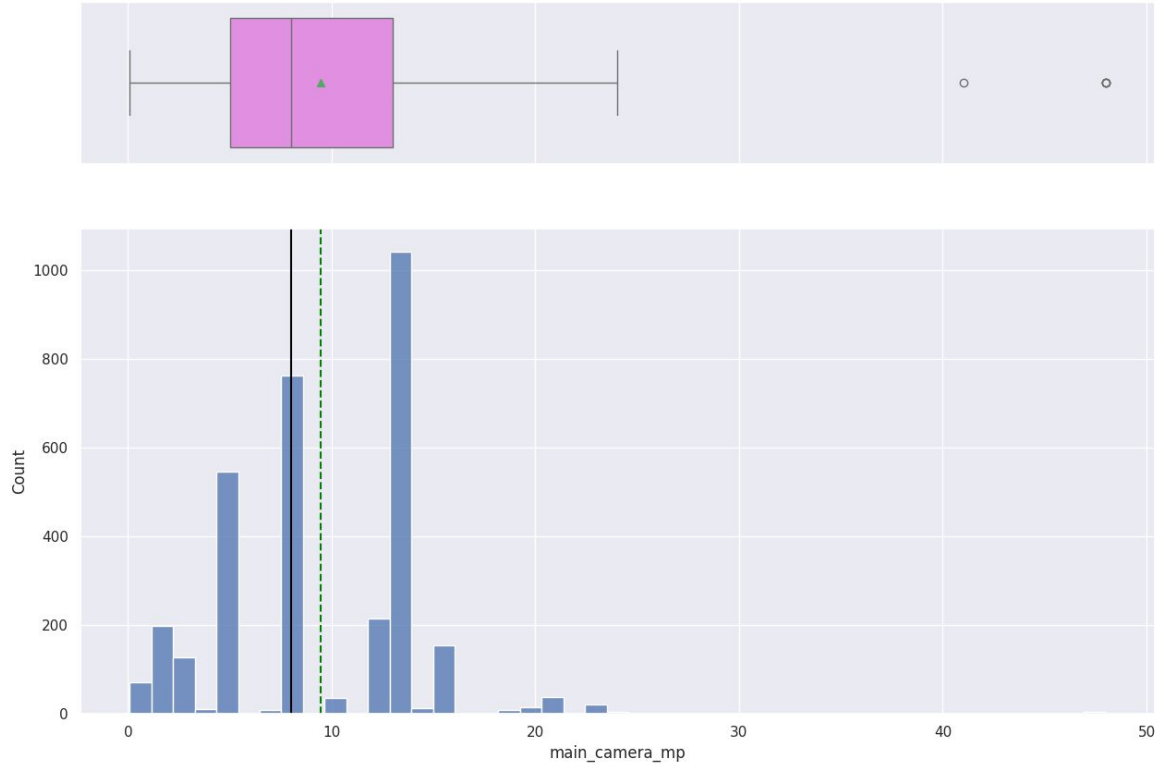


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EDA Results - Univariate- Main camera resolution

Main camera is generally between 5 to 15 Megapixel.

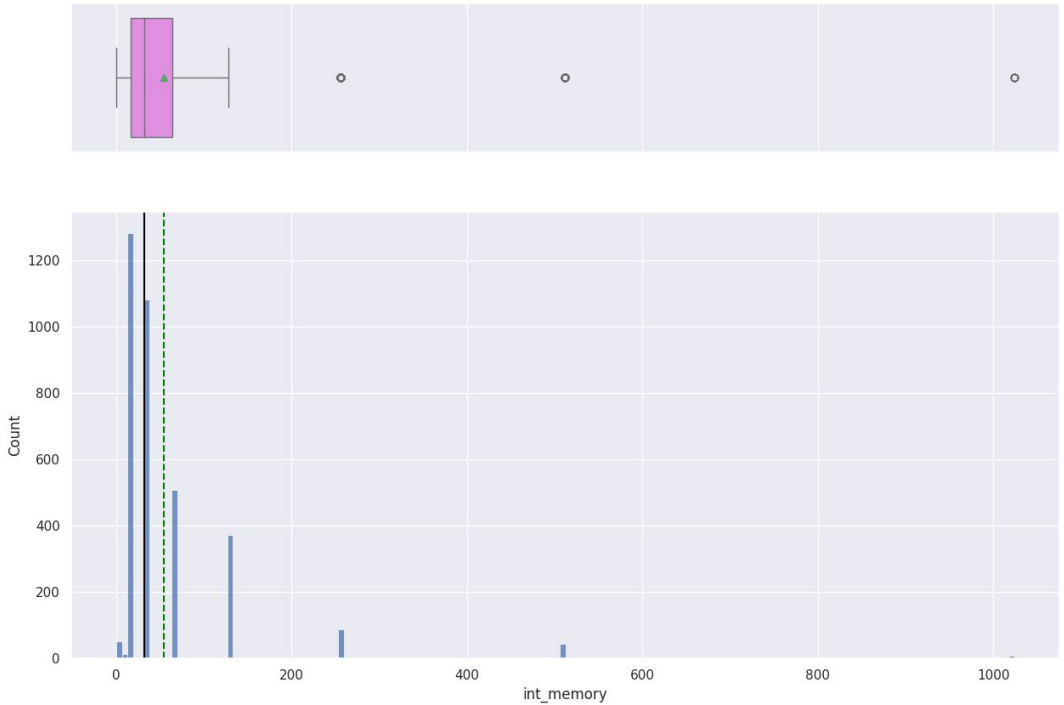


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EDA Results - Univariate - Internal Memory

There are some small number of specific memory configurations which are popular.
Internal memory is between 25 to 150 GB mostly.
Data looks right skewed.

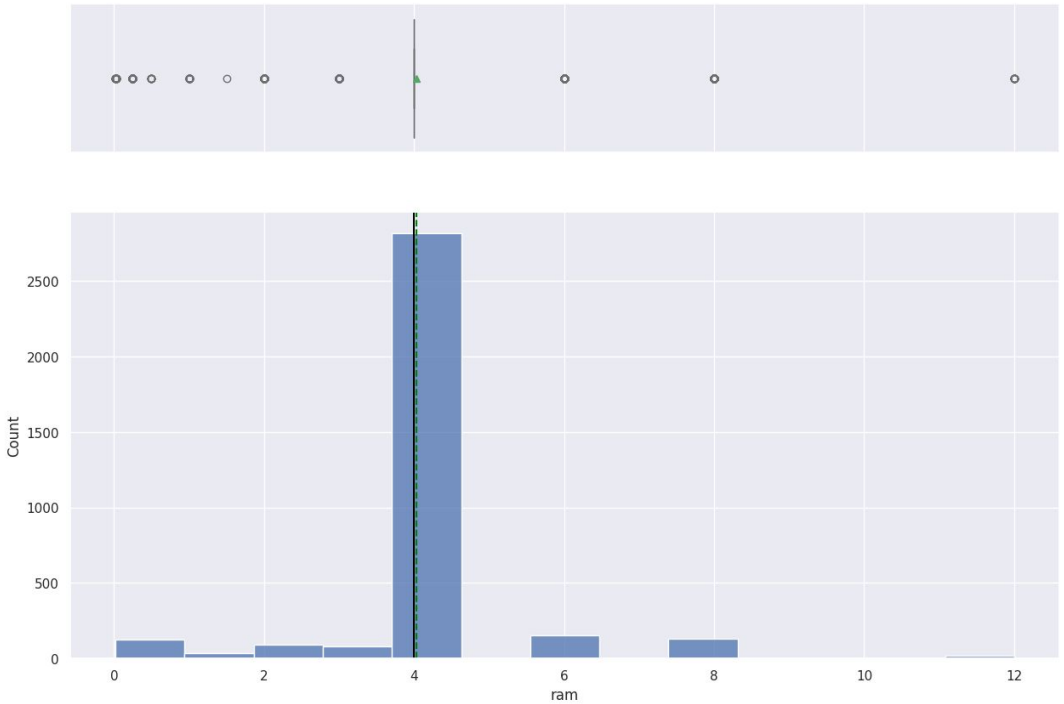


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EDA Results - Univariate - RAM

Most devices have RAM around 4GB.

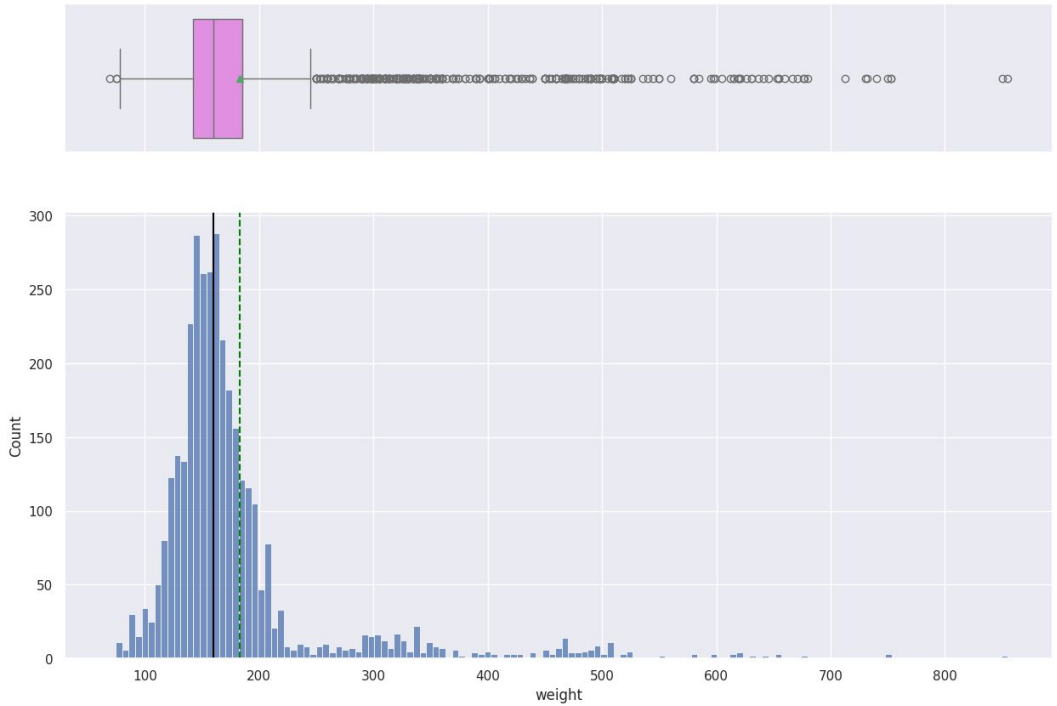


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EDA Results - Univariate - Weight

Weight looks right skewed. Most of the devices weigh between 100 to 200gm.

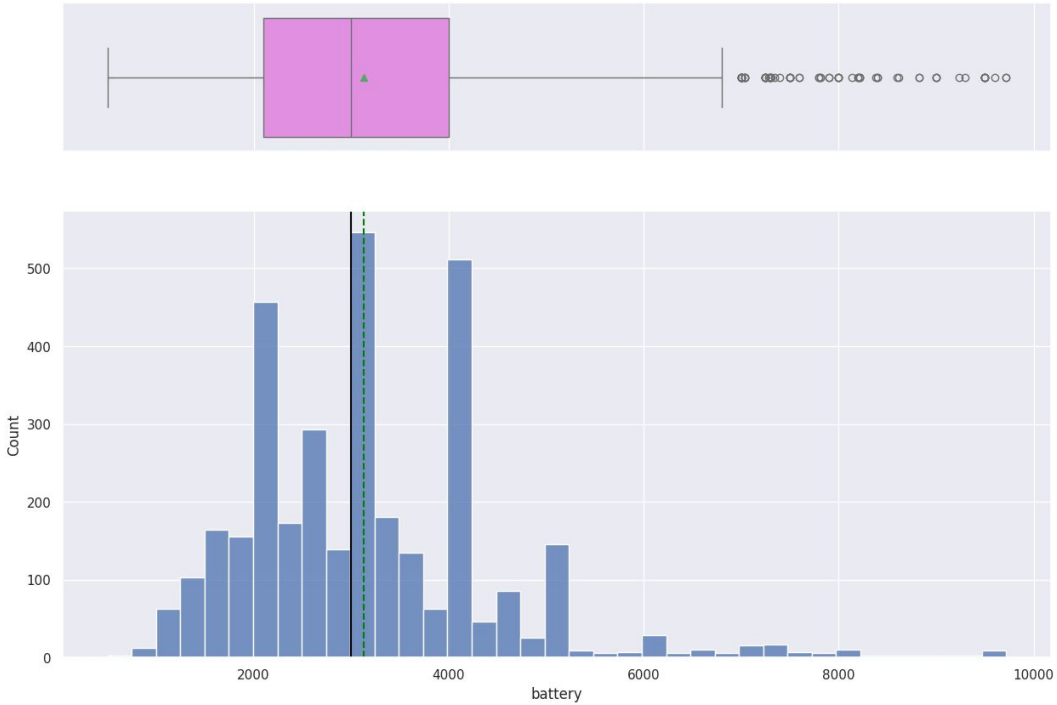


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EDA Results - Univariate - Battery

Battery life is mostly between 2000 to 4000 mAh.

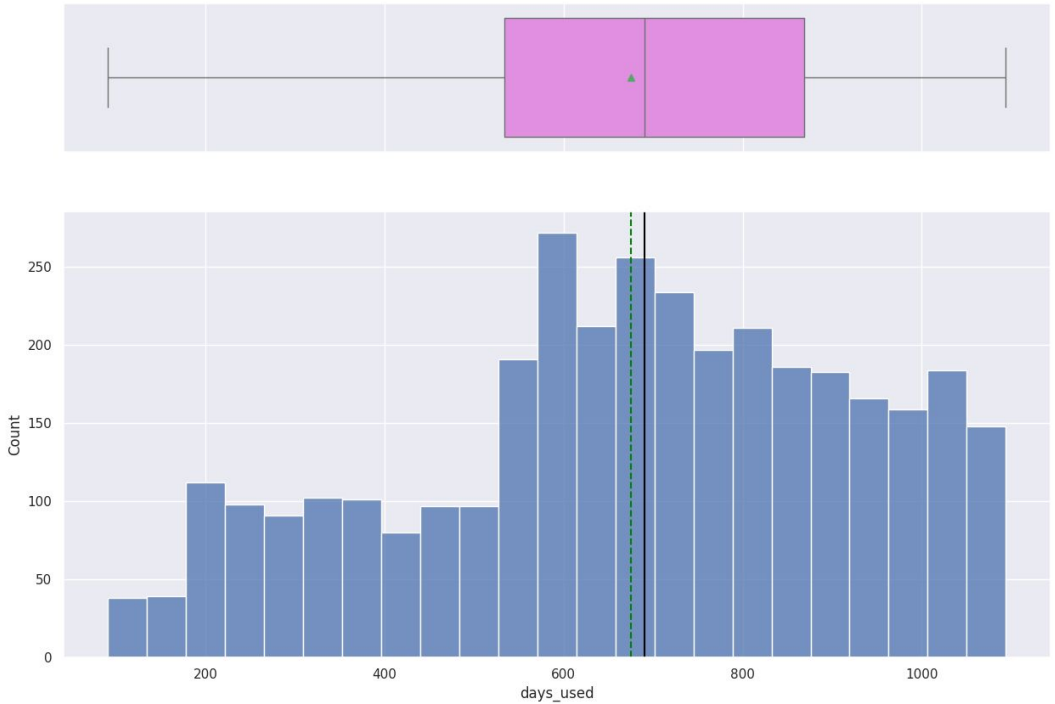


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EDA Results - Univariate - Days used

Most devices are used between 600 to 800 days.



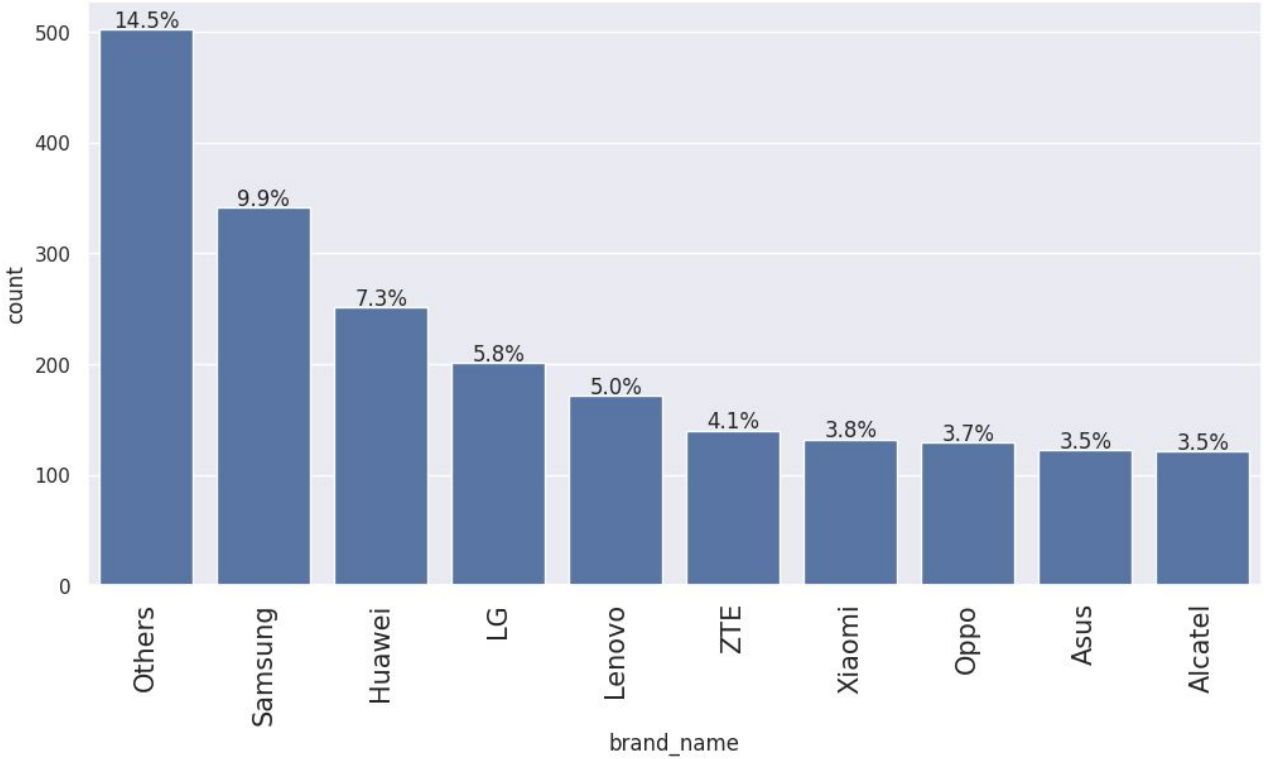
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EDA Results - Univariate - Brand name

Samsung is the top brand used.

Other brand names apart from the known brands make up 14. 5%

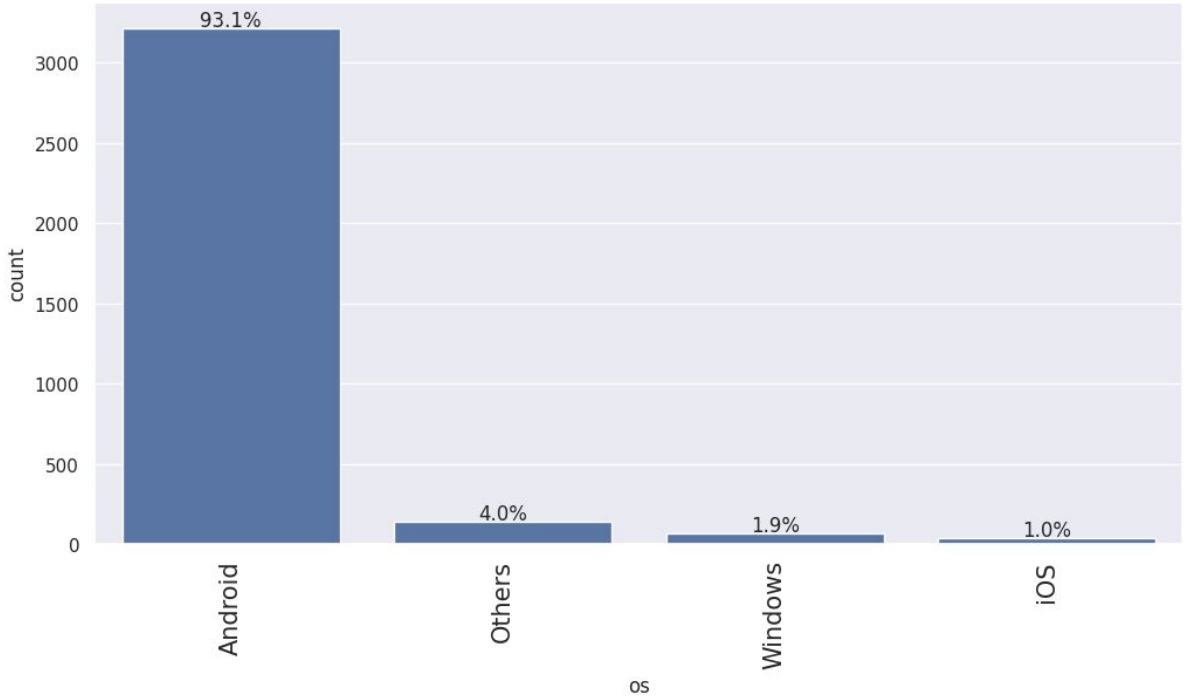


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EDA Results - Univariate - OS

Android is the most popular OS.



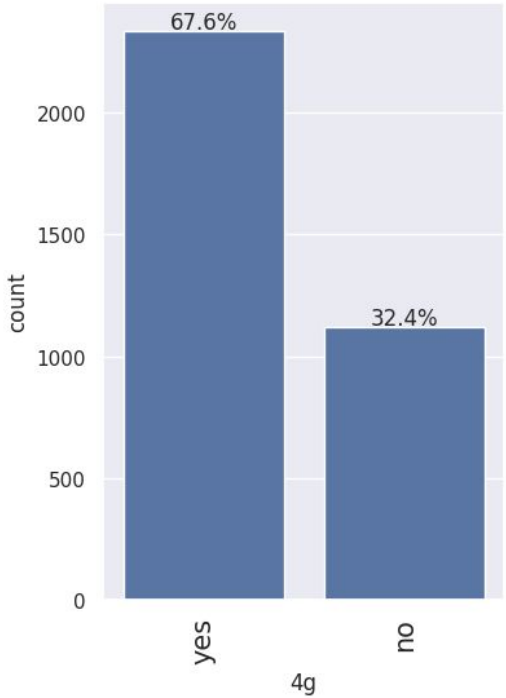
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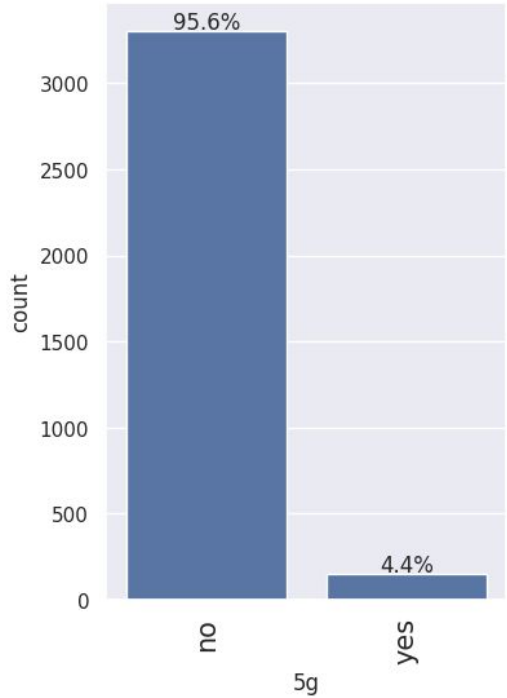
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EDA Results - Univariate - 4G and 5G

Most devices have 4g - 67.6%.

5g is only on 4.4% devices.





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EDA Results - Univariate - Release year

Devices released in 2014 were refurbished a lot.



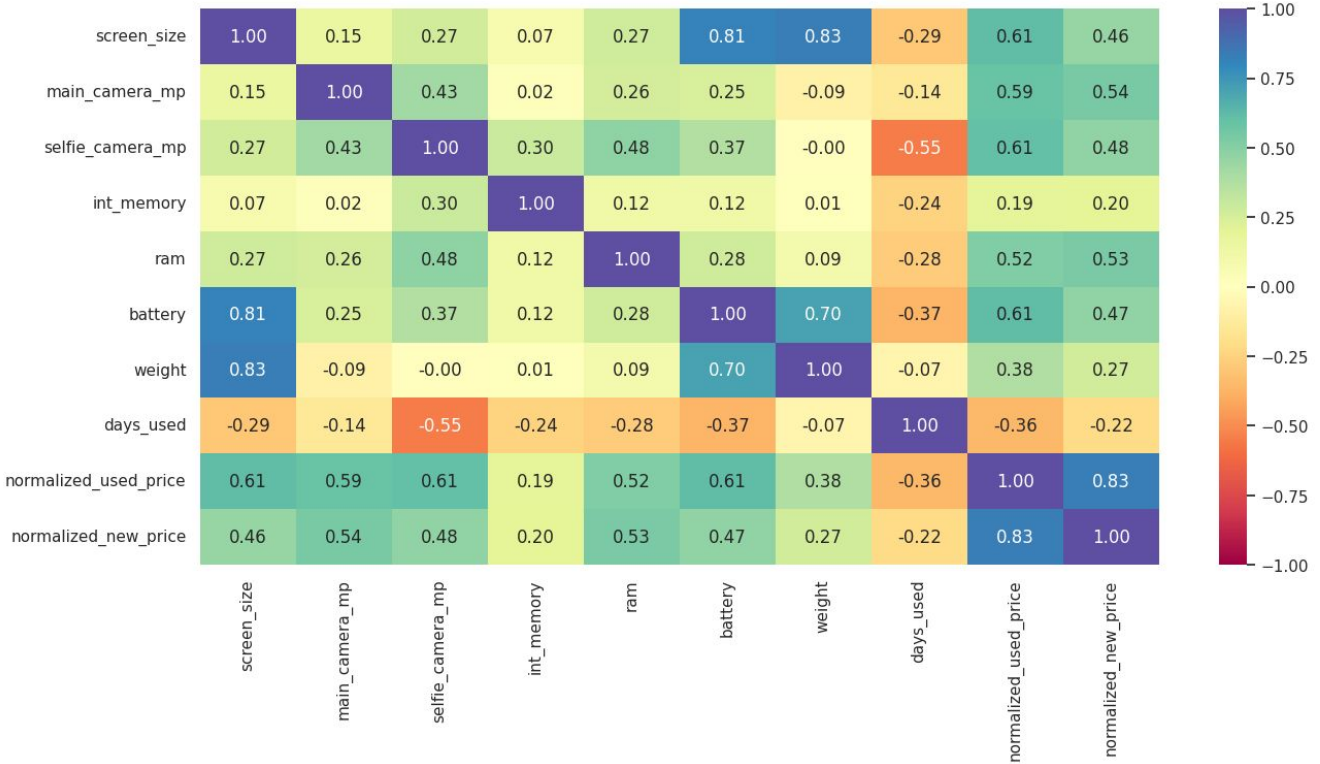
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EDA Results - Bivariate - Heat map

New price and used device price are correlated- which makes sense.

Screen size, battery and weight are correlated.

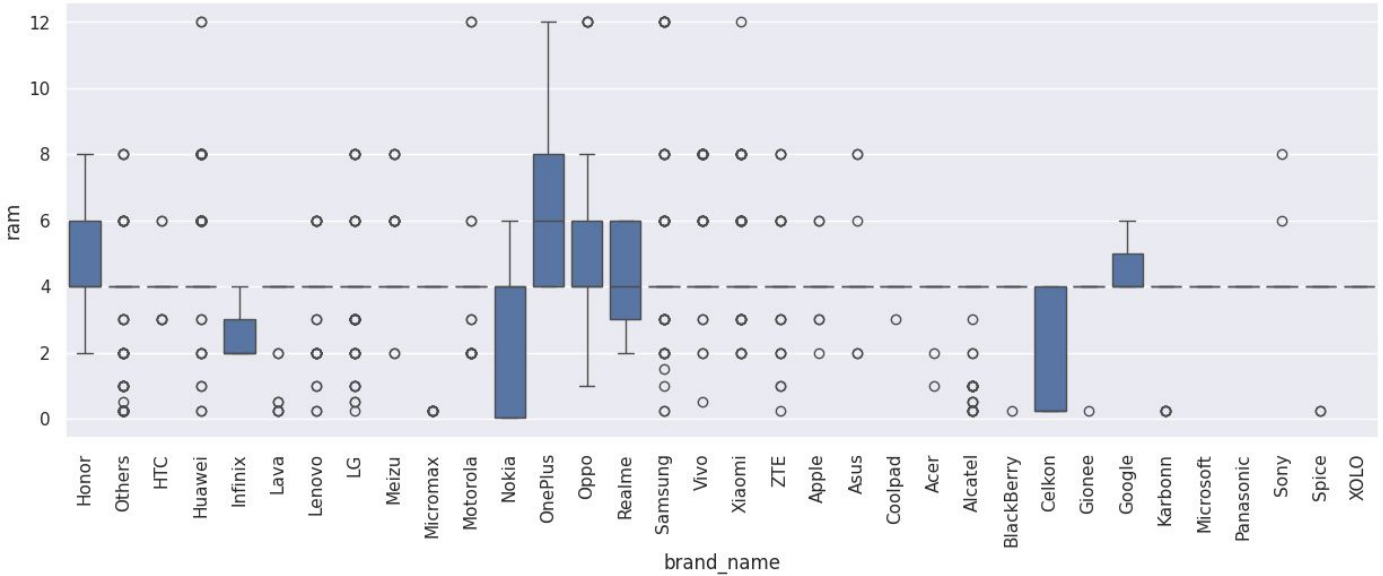


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EDA Results - Bivariate - RAM Vs Brand Name

Many brands have 4 GB as minimum RAM. Nokia and Celkon has maximum 4GB RAM.

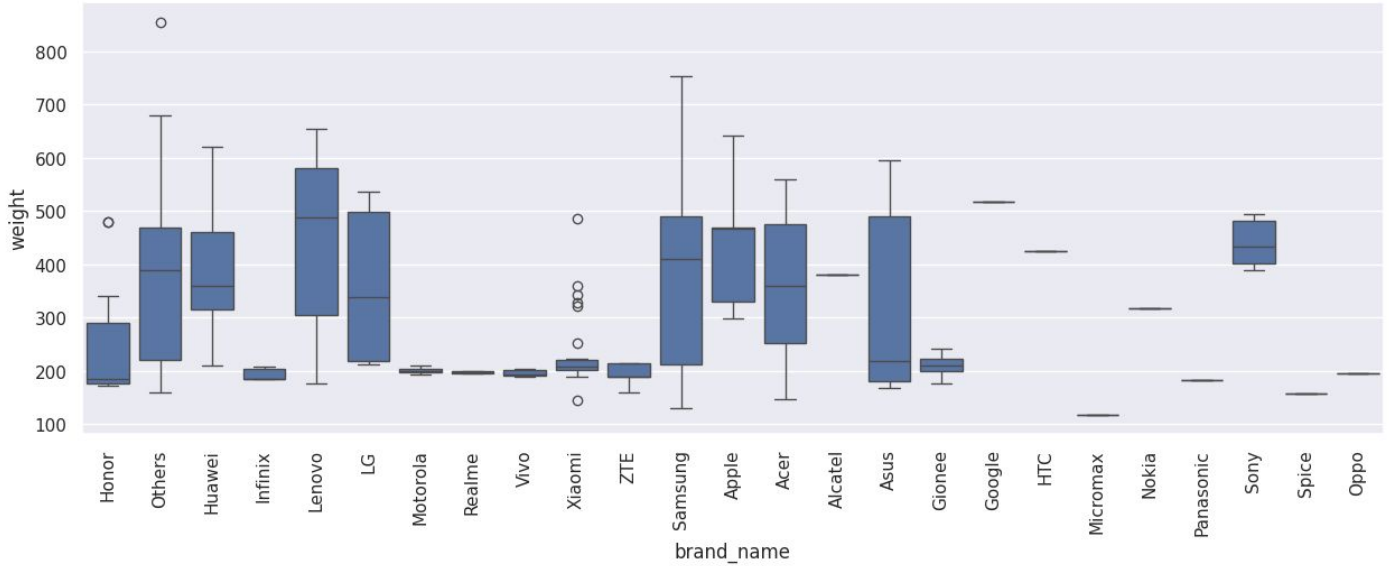


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EDA Results - Bivariate - Weight Vs Brand for large batteries

Most brands have weight between 200 to 500 gm

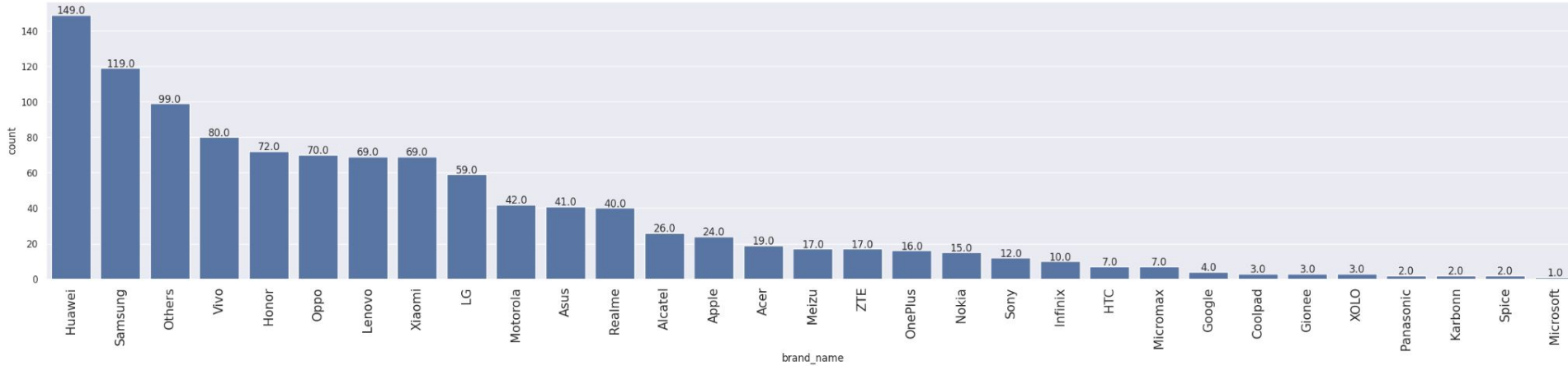


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EDA Results - Bivariate - Large screen phones

Other brands and Samsung have largest screens.

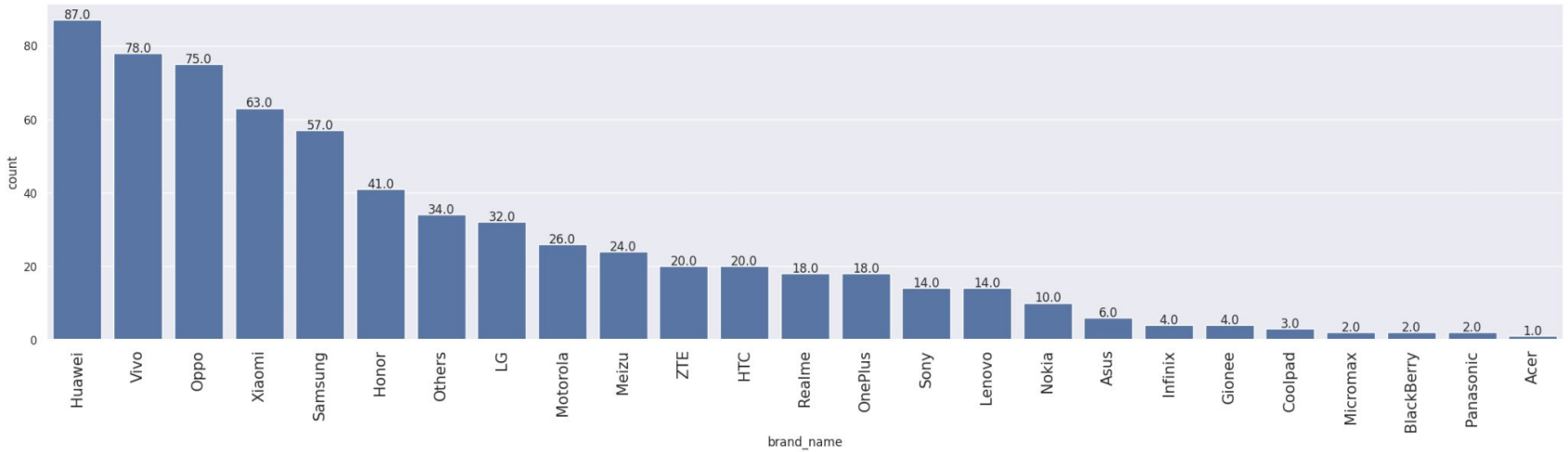


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EDA Results - Bivariate - Selfie cameras

Huawei has most devices with selfie cameras.

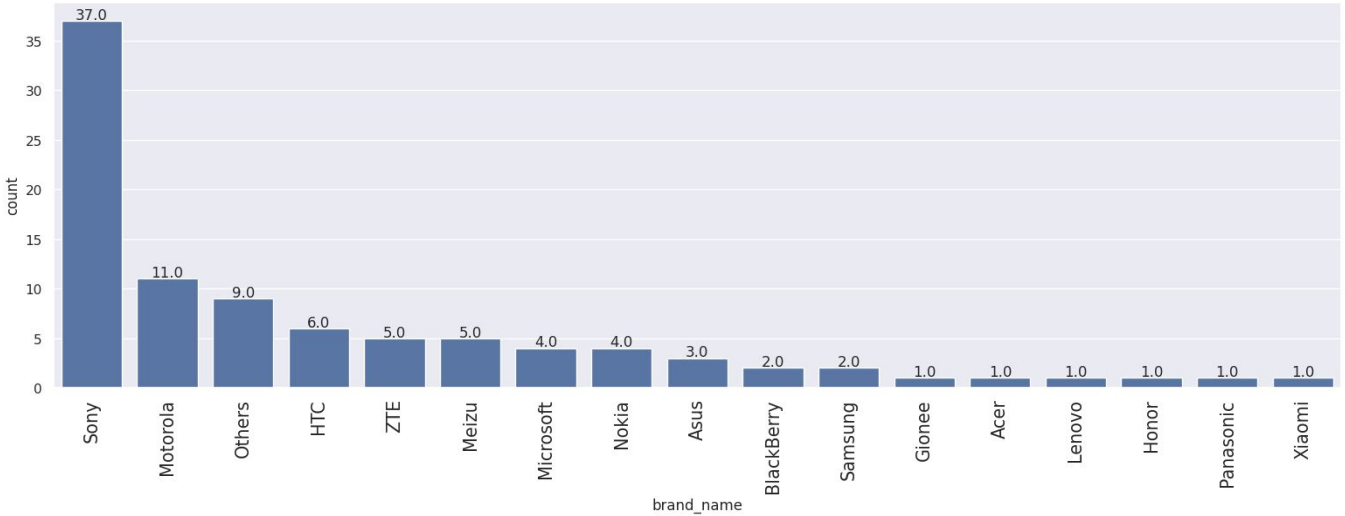


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EDA Results - Bivariate - Main cameras

Sony has most devices with main cameras.

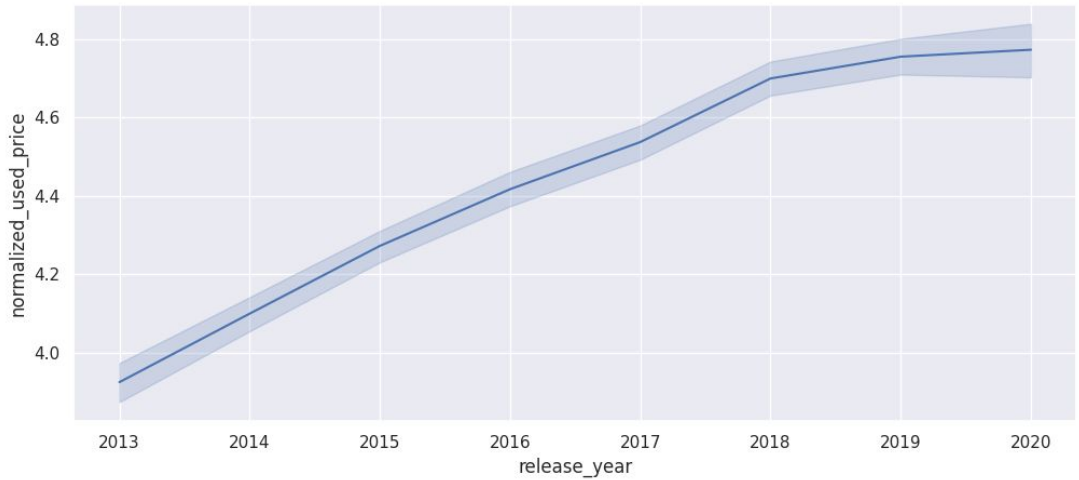


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EDA Results - Bivariate - Used price Vs Release year

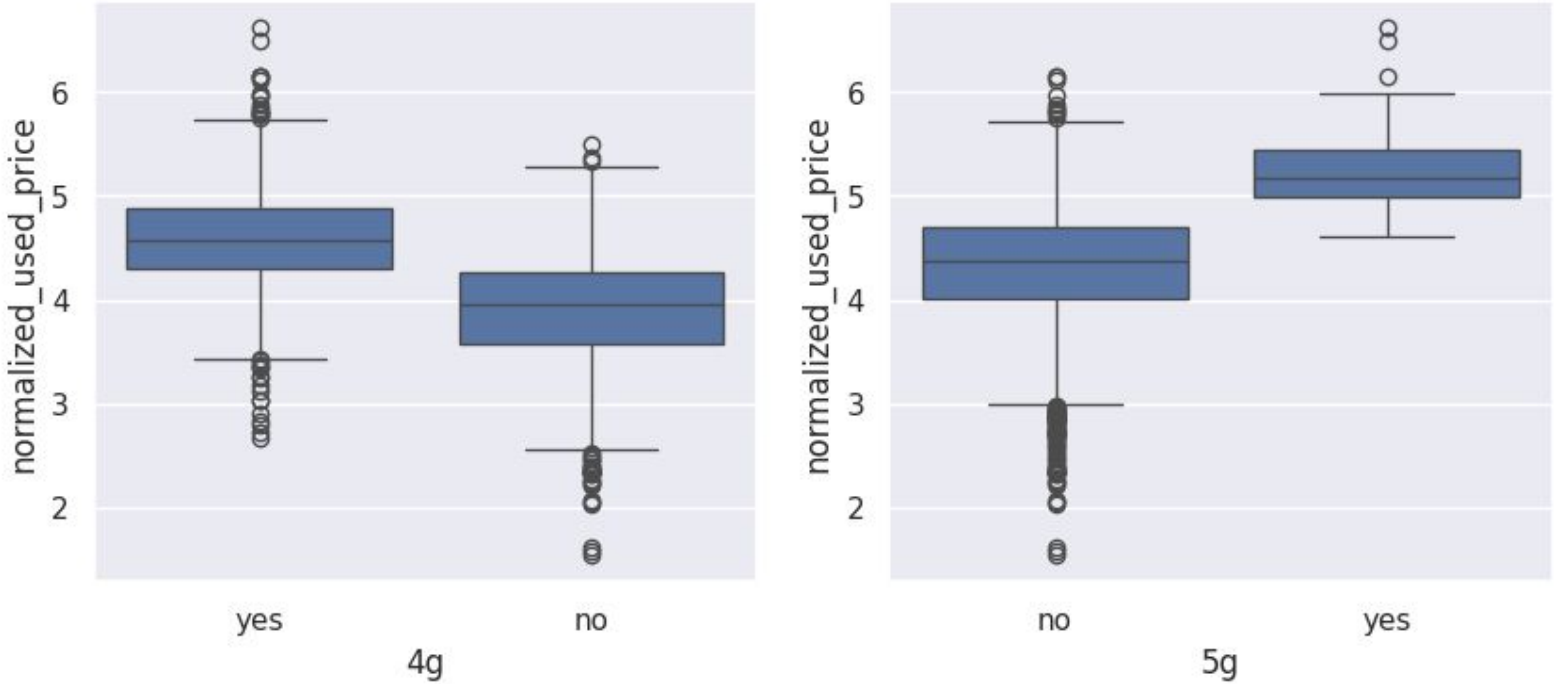
Used price increases with the release year increase.



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EDA Results - Bivariate - 4G and 5G prices



5g devices have a higher used price.

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Data Preprocessing- missing value imputation

main_camera_mp	179
selfie_camera_mp	2
int_memory	4
ram	4
battery	6
weight	7

main_camera_mp	**0**
selfie_camera_mp	**0**
int_memory	**0**
ram	**0**
battery	**0**
weight	**0**

Missing values counts in the data are shown as fixed in the second table.

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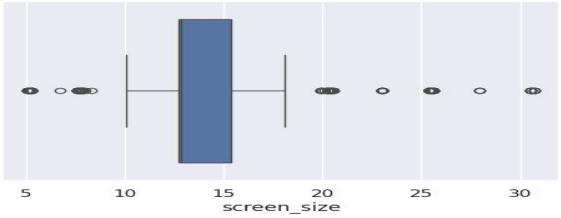
Data Preprocessing- Feature engineering

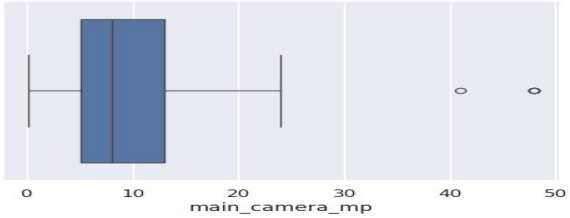
	years_since_release
count	3454.000000
mean	5.034742
std	2.298455
min	1.000000
25%	3.000000
50%	5.500000
75%	7.000000
max	8.000000

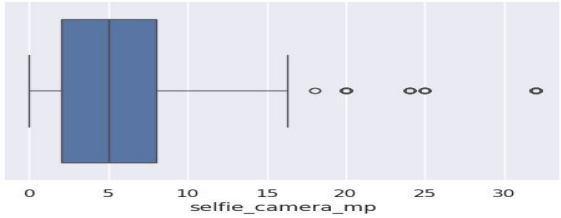
A new column was created called “years_since_release”. This will help in analysis.

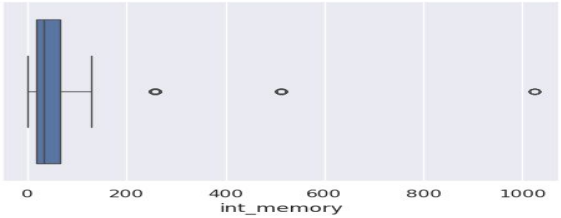
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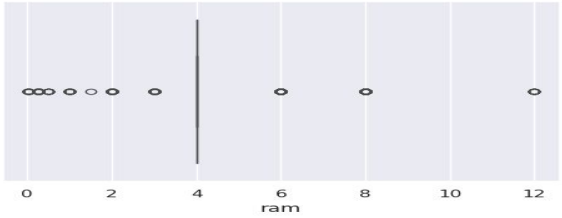
Data Preprocessing- Outlier check

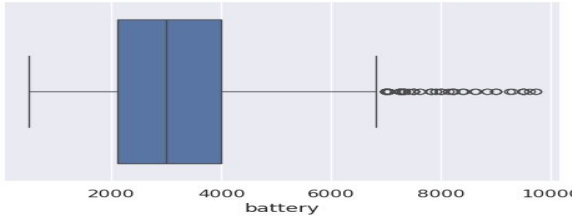


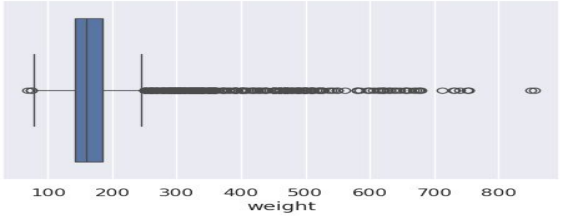




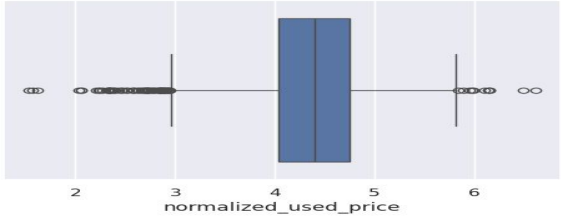




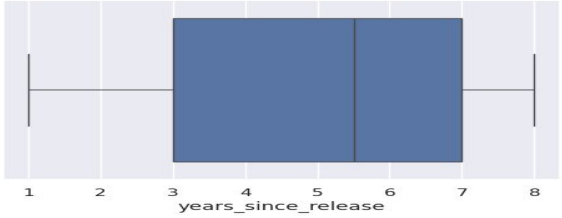












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Data Preparation - dropping normalised used price

```
brand_name       os  screen_size   4g   5g  main_camera_mp  \
0      Honor  Android        14.50  yes   no            13.0
1      Honor  Android        17.30  yes  yes            13.0
2      Honor  Android        16.69  yes  yes            13.0
3      Honor  Android        25.50  yes  yes            13.0
4      Honor  Android        15.32  yes   no            13.0

   selfie camera mp  int memory  ram  battery  weight  days used  \
0               5.0        64.0  3.0   3020.0   146.0        127
1              16.0       128.0  8.0   4300.0   213.0        325
2               8.0       128.0  8.0   4200.0   213.0        162
3               8.0        64.0  6.0   7250.0   480.0        345
4               8.0        64.0  3.0   5000.0   185.0        293

   normalized new price  years since release
0              4.715100                    1
1              5.519018                    1
2              5.884631                    1
3              5.630961                    1
4              4.947837                    1

0    4.307572
1    5.162097
2    5.111084
3    5.135387
4    4.389995
Name: normalized used price, dtype: float64
```

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Data Preparation - dummy variables added

```
Dummy boolean variables were added for all the categorical variables:
['brand name Alcatel', 'brand name Apple', 'brand name Asus',
       'brand_name_BlackBerry', 'brand_name_Celkon', 'brand_name_Coolpad',
       'brand name Gionee', 'brand name Google', 'brand name HTC',
       'brand name Honor', 'brand name Huawei', 'brand name Infinix',
       'brand_name_Karbonn', 'brand_name_LG', 'brand_name_Lava',
       'brand name Lenovo', 'brand name Meizu', 'brand name Micromax',
       'brand_name_Microsoft', 'brand_name_Motorola', 'brand_name_Nokia',
       'brand name OnePlus', 'brand name Oppo', 'brand name Others',
       'brand name Panasonic', 'brand name Realme', 'brand name Samsung',
       'brand_name_Sony', 'brand_name_Spice', 'brand_name_Vivo',
       'brand name XOLO', 'brand name Xiaomi', 'brand_name_ZTE', 'os_Others',
       'os_Windows', 'os_iOS', '4g_yes', '5g_yes']
```

Data was divided into 70:30 ratio for training:testing

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Data Preparation - Model performance summary

OLS Regression Results

Dep. Variable:	normalized_used_price	R-squared:	0.839
Model:	OLS	Adj. R-squared:	0.838
Method:	Least Squares	F-statistic:	963.5
Date:	Sun, 20 Jul 2025	Prob (F-statistic):	0.00
Time:	06:51:48	Log-Likelihood:	79.056
No. Observations:	2417	AIC:	-130.1
Df Residuals:	2403	BIC:	-49.05
Df Model:	13		
Covariance Type:	nonrobust		

84 percent of variance was explained by this model.
There are 13 predictors.
This model is OK for prediction.

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Data Preparation - Training and test performance

Training performance

	RMSE	MAE	R-squared	Adj. R-squared	MAPE	
0	0.234184	0.18308	0.839028	0.83809	4.403666	

Test performance - Data is comparable to training. Model is good.

	RMSE	MAE	R-squared	Adj. R-squared	MAPE	
0	0.241747	0.186868	0.837967	0.835748	4.562763	

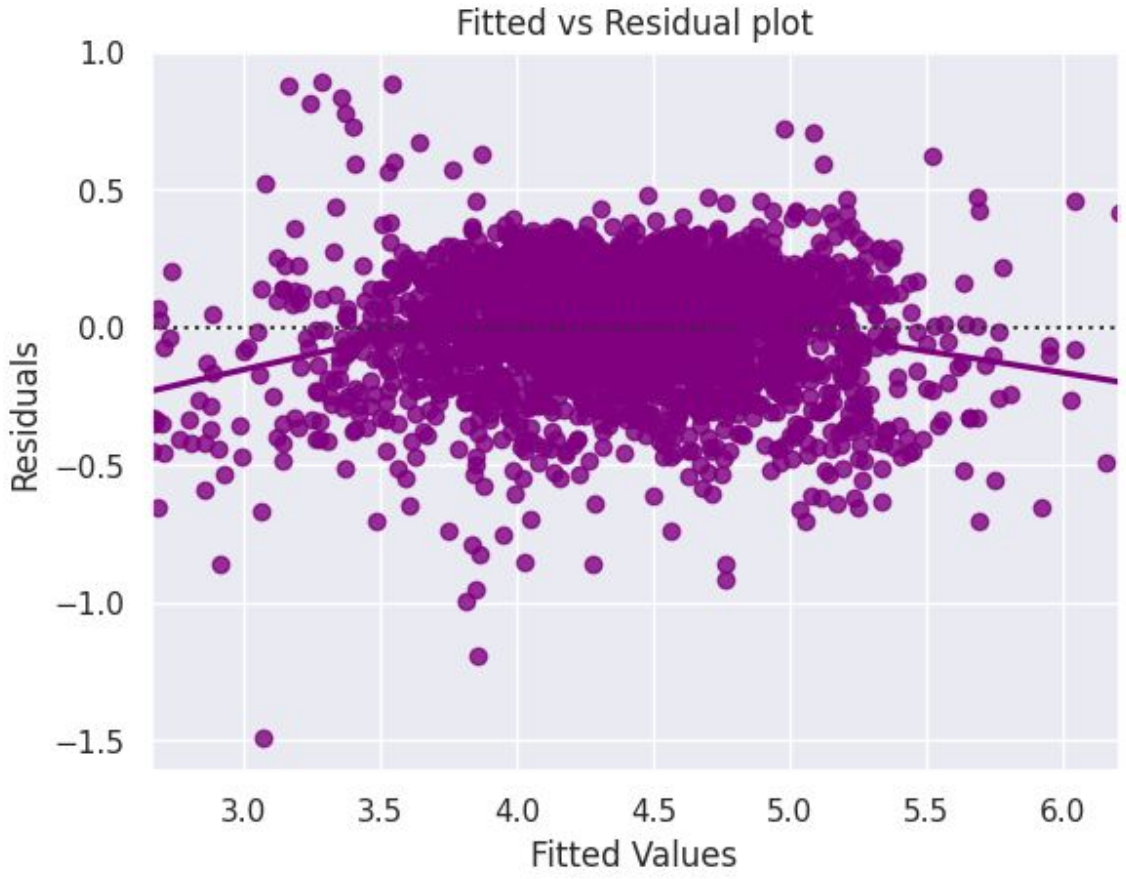
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Data Preparation - Multicollinearity

- There were many columns with high VIFs
- Multicollinearity was present and had to be treated.
- Did not choose to drop the dummy variable columns as this was only a single variable before, so this would alter the data.
- Os_ios and screen_size columns were dropped for highest VIF values.
- High P values were also dropped.
- After this, 13 predictors were obtained. High VIFs were not present anymore at this stage.

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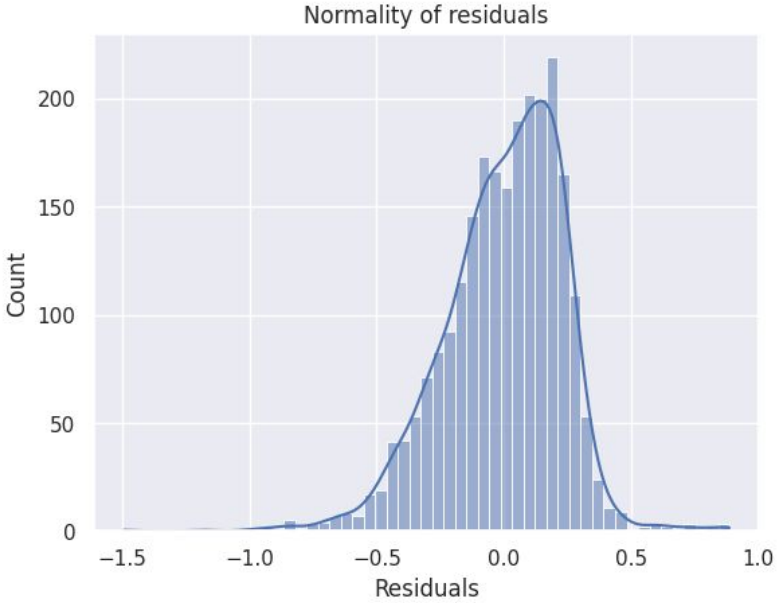
Test for linearity and independence

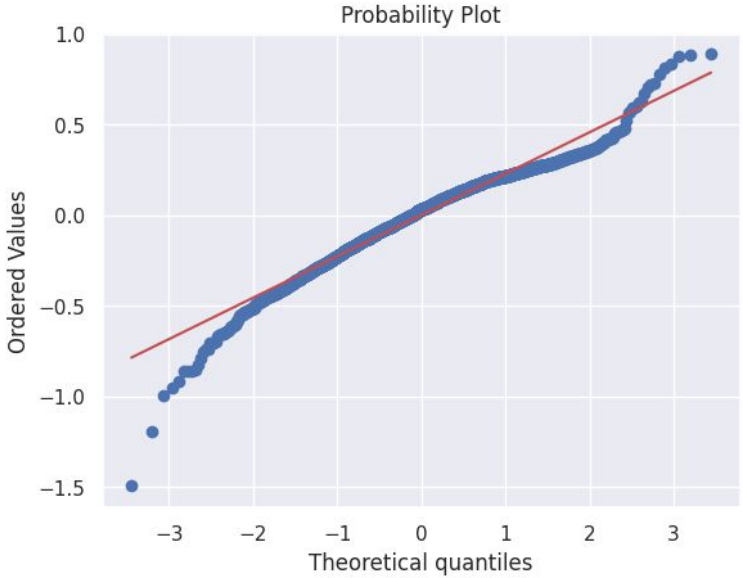


There is no pattern in the data. Variables are independent.

ıibited.

Test for normality





Residuals are normally distributed. The model also passed the Shapiro test.
Residuals are also homoscedastic.

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Final model results

OLS Regression Results

Dep. Variable:	normalized_used_price	R-squared:	0.839
Model:	OLS	Adj. R-squared:	0.838
Method:	Least Squares	F-statistic:	963.5
Date:	Sun, 20 Jul 2025	Prob (F-statistic):	0.00
Time:	07:21:39	Log-Likelihood:	79.056
No. Observations:	2417	AIC:	-130.1
Df Residuals:	2403	BIC:	-49.05
Df Model:	13		
Covariance Type:	nonrobust		

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